Κάρυος Γεώργιος - Lab 02 - 30738

Εισαγωγή στην Βιοστατική & Στατιστική Ανάλυση δεδομένων, Βιοϊατρικά Συστήματα Υγείας

2.1)

Επάγγελμα	fi	σχετική fi	%
Ιδιωτικός	6	6/20 = 0.3	30%
Δημόσιος	7	0.35	35%
Αυτοαπασχ	5	0.25	25%
Άλλο	2	0.10	10%

σύνολο: N=20

ιδιωτ. $f=6 \Rightarrow F/N = 6/20 = 30\%$
Δημ. $7/20 = 0.35 = 35\%$
Αυτ. $5/20 = 25\%$
Άλλο $2/10 \Rightarrow 10\%$

} 100%

2.3) Συνολο κλήσεων 60400

- Φωτιές: $47\% \Rightarrow 60400 \cdot 0,47 = 28388$
- Ανεμοστρόβιλοι: $27\% \Rightarrow 60400 \cdot 0,27 = 16308$
- Βοήθεια: $27\% \Rightarrow 60400 \cdot 0,21 = 12684$
- Ψευδείς: $5\% \Rightarrow 60400 \cdot 0,05 = 3020$

} συνολο 60400

2.4) N=80: πτυχιούχοι ΑΕΙ: $20\% \Rightarrow 80 \cdot 0,20 = 16$
— ΤΕΙ: $50\% \Rightarrow 80 \cdot 0,5 = 40$
Λύκειο: $30\% \Rightarrow 80 \cdot 0,3 = 24$

} συνολο 80 πτυχ

2.9) 1) Αετρέας: 15, 18, 18, 17 $\quad X_A = \frac{15+18+18+17}{4} = 17$

2) Βασίλης (+2): 17, 20, 20, 19 $\quad X_B = \frac{17+20+20+19}{4} = 19$

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3) Γιαννης (-4): 11, 14, 14, 13 $\quad X_{\Gamma} = \frac{11+14+14+13}{4} = 13$

2.12) α) 1, 2, 6 $\quad x = \frac{7+2+6}{3} = 3$ / διαμεσος = 2

β) 2, 4, 12 $\quad x = \frac{2+4+12}{3} = 6$ / διαμεσος = 4

γ) 11, 12, 16 $\quad x = \frac{11+12+16}{3} = 13$ / διαμεσος = 12

δ) 12, 14, 22 $\quad x = \frac{12+14+22}{3} = 16$ / διαμεσος 14

οπου x = μεση τιμη

2.13) τιμές: (mA) : 27.5, 22.7, 21.3, 27.7, 22.0, 22.2, 21.8, 21.4, 21.9, 22.7,

αθροισμα : sum ολων των πιο πανω = 218.0

X (μεση τιμη) = $\frac{218}{10} = 27.8$

Τυπικη Αποκλιση : $s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}} \simeq 0,316$

ΑΣΚΗΣΕΙΣ ΣΤΑΤΙΣΤΙΚΗΣ

Ασκ

1. • N = 50 , sum = 2290
   • Μεσος ορος: Y = 2290 / 50 = 45.8
   • Διαμεσος: 25η θεση = 26η θεση = 45 $\Rightarrow \frac{90}{2} = 45$
   • Επιμερα τουσα τιμη → 45 (7 φορες)
   • min = 17 , max = 86
   • Ευρως = 86 - 17 = 69 nice

   • Διακυμανση: $\sum (y_i - \bar{y})^2 = 14372.2 \Rightarrow s^2 = \frac{14372.2}{49} = 293,31$

   τυπικη αποκλιση: $\sqrt{293,31} = 17.13$

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Πίνακας, για κάθε βαθμό $Y_i - \bar{Y}$ & $(Y_i - \bar{Y})^2$

φ/λ	Y_i	$Y_i - 45,8$	$\wedge 2$
1	17	$17-45,8=-28,8$	829.44
2	18	$-27,8$	772.84
3	22	$-23,8$	566.44
4	24	$-21,8$	475.24
5	25	$-20,8$	432.64
..	...	...	...
46	76	30,2	912.04
47	79	33,2	1102.24
48	80	34,0	1169.64
49	83	37,2	1383.84
50	86	40,2	1616.04

$\Rightarrow \sum Y_i = 2290,\ \sum (Y_i - 45,8) = 0,\ \sum (Y_i - 45,8)^2 = 14372.2$

Ιστογράμμα: φοιτητών
10-20 → 2φ, 27-30 → 6φ, 31-40 → 8φ, 47-50 → 17φ
51-60 → 4φ, 61-70 → 6φ, 71-80 → 5φ, 87-90 → 2φ

αση 2. α) Συνόλο Ανδρών : $N = 5387$

β) πράσινα μάτια & ξανθια μαλλια: 343
$(343 / 5387) \cdot 100 = 6,37\%$

γ) απο γαλάζια μάτια (1580), ανοιχτό καφέ (584)
$(584/1580) \cdot 100 = 36.96\%$

δ) απο κοκκινα μαλια (286), καφετιζια (48)
$(48/286) \cdot 100 = 16.78\%$

ε) 95% ΔΕ για αναλογια γαλάζια+ξανθια
$p = 1688/5387 = 0,1277$

ς) Σε 80000 $\Rightarrow 80000 \cdot 0,1277 = 10216$

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ασκ 3

- sum = 1710
- Μεσος ορος : $x = 1710 / 12 = 142.5$
- διαμεσος: $(142 + 143)/2 = 142,5$

εχουμε $SS = 143$

$\Rightarrow s^2 = (143/11) = 13$

$s = \sqrt{13} = 3,67$

$SE = s/\sqrt{12} = 3,67 / 3,43 = 7.04$

$CV = (s/\bar{x}) \cdot 100 = \frac{3,67}{142.5} \cdot 100 = 2,53\%$

Και είναι ομοιογενές διοτί $CV < 10\%$

Ασκ4

πρωτεινη, $N = 14$

sum = 165,8 , $MO \Rightarrow \bar{x} = 165,8/14 = 11,84$

διαμεσος: $(11,8 + 12,0)/2 = 11,9$

sum τετραγωνων διαφορων : $SS = 9,554$

$s^2 = (9,554/13) = 0,735$

$s = \sqrt{0,735} = 0,857$

$SE = 0,857/\sqrt{14} = 0,229$

$CV = 0,857/11,84 \cdot 100 = 7,24\%$

ομοιογενειες $CV < 10\%$

Ασκ5

NA: dead 45, alive 15 $\Rightarrow$ total = 60

% νεκρων (NA) $= (45/60) \cdot 100 = 75\%$

BG dead 25, alive 35 $\Rightarrow$ total = 60

% ζωντανων (BG) $= (35/60) \cdot 100 = 58,33\%$

Συνολο πεθαμενων 45 + 25 = 70

απο τα νεκρα ; % με NA

$(45/70) \cdot 100 = 64,28\%$

MATLAB

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Georgios

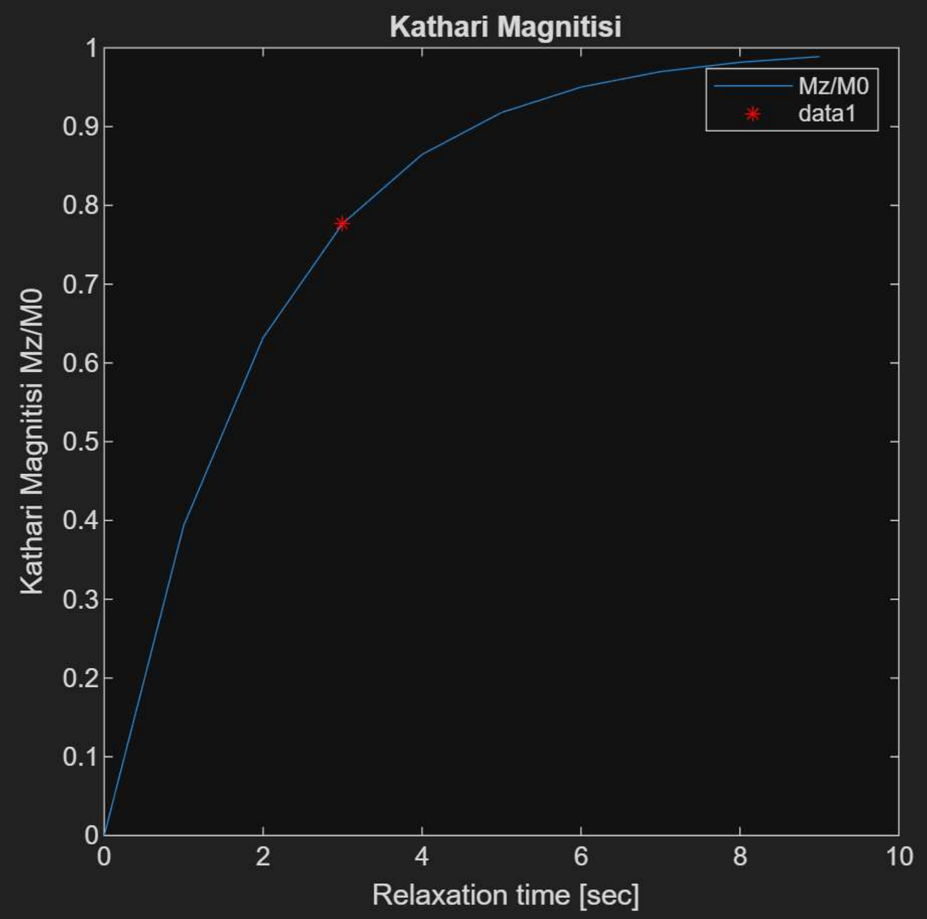
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```
%Exercises Chapter 3:
%Ex3-5:
%Enas istow echi kathari magnitisi T1=2 sec. An i T1 mideniszotan, poso chrono tha pari
%gia tin kathari magnitiosi na anaktisi to 98% tiw
disp('Chronos pou chriazete in kathari magnititsi gia na pai sto 98% tiw '), pause(0.01);
%Draw the M=Mz/M0
t = [0:1:9]; T1 = 2.0 *(ones(1,10));
M = ones (1, 10)-exp(-t./T1);
%f = input('Enter the frequency of sound f in HZ:  ');
%Plot the M vs t
%Find the 98% of the hight for a specific time
time = round (-1.0 * log(1-0.98) );
M_at_time  = M(time-1);
fprintf('The Magnetizion M : %6.8f mm for a time of %2.1f secs\n', M_at_time, time);
figure, plot (t, M); title(' Kathari Magnitisi');
xlabel (' Relaxation time [sec]'); ylabel ('Kathari Magnitisi Mz/M0');
legend ( 'Mz/M0'), hold on, plot(time-1, M(time), 'r*');
```

Figure 1



Command Window

```
>> untitled
Chronos pou chriazete in kathari magnititsi gia na pai sto 98% tiw
The Magnetizion M : 0.63212056 mm for a time of 4.0 secs
>> ✨ Press Ctrl + Shift + P to generate code with Copilot
```

Editor: 100% UTF-8 CRLF Script Ln 17 Col 58

17°C Sunny

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9:49 am 19/02/2026

MATLAB

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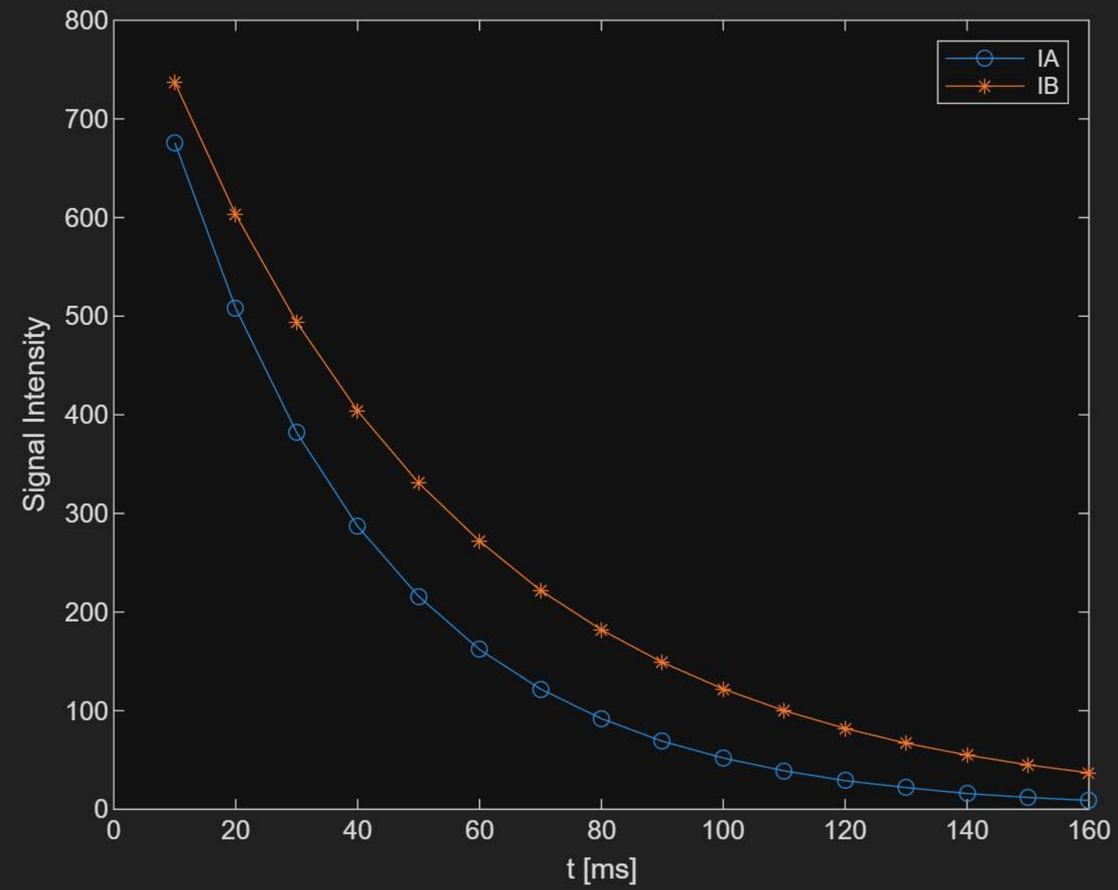
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```
%Ex.3-7:
%Given the following values plot, (i) the signal intesnisty versus
%time for the two diffeernt tissues.
t=[10:10:160];
Ia=[676;508;382;287;216;162;122;92;69;52;39;29;22;16;12;9];
Ib=[737;603;494;404;331;272;222;182;149;122;100;82;67;55;45;37];
figure(1), plot(t, Ia, '-o'), xlabel ('t [ms]'); ylabel ('Signal Intensity');
hold on, plot(t, Ib, '-*'),
legend ( ' IA ', ' IB ' ),
hold off;
%Find the T2 for A and B tissues
TA2 = (37*max(Ia))/100;
TB2 = (37*max(Ib))/100; %plot( t(1),Ia(1), 'r*'); plot( t(1),Ib(1), 'b*');
fprintf('The time for tissue A is: %5.2f secs and the time for the tissue B is: %5.2
%Find the contrast between the two tissues and plot C=(a-B)/noise;
noise = input('Enter the noise : ');
%Calculate CNR
CNR = (Ia-Ib)/ noise;
% Plot the contrast versus time
figure(2), plot(t, abs(CNR), '-o'), xlabel ('t [ms]'); ylabel ('CNR');
%Find the maximum of the contrast
maximum = max(abs(CNR));
fprintf('The maximum of the contrast is:  %5.2f \n', maximum);
```

Figure 1 Figure 2



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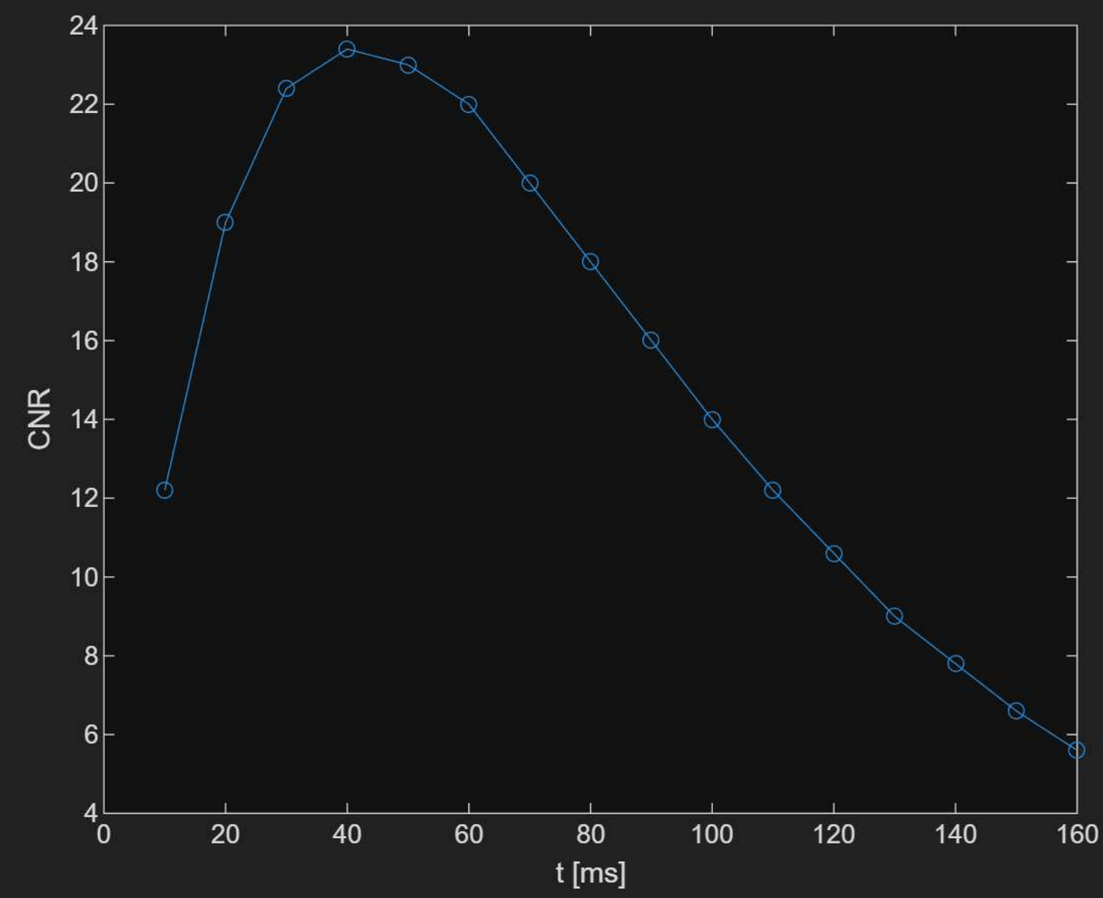
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```
1   %Ex.3-7:
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3   %time for the two diffeernt tissues.
4   t=[10:10:160];
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6   Ib=[737;603;494;404;331;272;222;182;149;122;100;82;67;55;45;37];
7   figure(1), plot(t, Ia, '-o'), xlabel ('t [ms]'); ylabel ('Signal Intensity');
8   hold on, plot(t, Ib, '-*'),
9   legend ( ' IA ', ' IB ' ),
10  hold off;
11  %Find the T2 for A and B tissues
12  TA2 = (37*max(Ia))/100;
13  TB2 = (37*max(Ib))/100; %plot( t(1),Ia(1), 'r*'); plot( t(1),Ib(1), 'b*');
14  fprintf('The time for tissue A is: %5.2f secs and the time for the tissue B is: %5.2
15  %Find the contrast between the two tissues and plot C=(a-B)/noise;
16  noise = input('Enter the noise : ');
17  %Calculate CNR
18  CNR = (Ia-Ib)/ noise;
19  % Plot the contrast versus time
20  figure(2), plot(t, abs(CNR), '-o'), xlabel ('t [ms]'); ylabel ('CNR');
21  %Find the maximum of the contrast
22  maximum = max(abs(CNR));
23  fprintf('The maximum of the contrast is:  %5.2f \n', maximum);
```

Figure 1 | Figure 2



Command Window

```
>> untitled
The time for tissue A is: 250.12 secs and the time for the tissue B is: 272.69 secs
Enter the noise : 5
The maximum of the contrast is:  23.40
>>
```

17°C Sunny

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9:59 am 19/02/2026